

# The Icosahedral Polyhedron Computer: A Hierarchical Lattice Architecture for Embodied Superintelligence Powered by the HSMT Meta-Transistor Engine

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## Abstract

This paper presents the Icosahedral Polyhedron Computer, a three-dimensional hierarchical lattice architecture in which every vertex of a self-similar icosahedral framework hosts a fully self-contained Meta-Transistor Engine. The design integrates the Meta-Transistor Engine with the Compound Multi-Logic Engine (CMLE), which implements a progressive logic system extending from binary and balanced ternary through volumetric spherical logic and native hypercomplex operations. The CMLE further incorporates fractal self-similar logic layers, thermodynamically-aware computation, and an explicit observer/measurement layer.

The architecture employs nine-layer hybrid Carbon Nano-Onions interconnected through Radial Multi-Bus Nanowires (RMBNs) and Inter-Node Coaxial Nanowires (INCNs), following a tapering allocation of  $12 \rightarrow 8 \rightarrow 8 \rightarrow 6 \rightarrow 6$  channels per node with a strict 6/6 separation of power and data pathways. This creates a natural gradient of increasing resonance quality (graduated quietness) toward the center while supporting high-efficiency hypercomplex and fractal computation.

Extensive radial-lattice simulations establish concrete performance baselines. Binary gates achieve 0.40–0.55 aJ per operation and ternary gates 0.75–1.05 aJ, with tier-transition success rates above 98%. The architecture is a classical resonant system grounded in Hierarchical Shell-Manifold Theory and provides a coherent, scalable, and energy-efficient pathway toward safe and embodied superintelligence.

## 1 Introduction

The Hierarchical Shell-Manifold Theory (HSMT) provides a unified geometric framework in which spacetime is organised into integer-indexed radial shells coupled by a tunable Gaussian projection kernel and governed by a single self-adjoint octonionic Master Operator. The Meta-Transistor Engine represents the engineered realisation of this framework: a high-Q resonant chamber capable of dynamically modulating the local blurriness field to control inter-shell leakage.

This architecture enables the Compound Multi-Logic Engine (CMLE), a progressive, multi-dimensional logic system that evolves from binary and balanced ternary through volumetric spherical logic and native hypercomplex operations. The CMLE further incorporates fractal self-similar logic layers that operate recursively across radial scales, thermodynamically-aware computation that distinguishes reversible and irreversible operations, and an explicit observer/measurement layer that formalizes context-dependent readout and reflective computation.

The Icosahedral Polyhedron Computer arranges these resonant chambers within a self-similar, hierarchical icosahedral lattice. The fundamental repeating unit consists of one central carbon nano-onion (CNO) surrounded by twelve outer CNOs. These units employ a graduated, dual-mode connectivity scheme in which the outer radial zones utilise full level-to-level nanowire implantation, the middle zones adopt a hybrid approach, and the inner core zone maintains minimal penetration to preserve high resonance quality. This architecture creates a natural gradient of increasing signal stability and control authority toward the center (graduated quietness).

This architecture constitutes a classical resonant computing system that achieves high information density and rich representational capacity through multi-valued logic, continuous phase encoding, fractal self-similar reasoning, and native quaternionic and octonionic algebraic structures, without any requirement for quantum coherence. This paper presents the complete specification of the Icosahedral Polyhedron Computer, details the Radial Multi-Bus Nanowire (RMBN) and Inter-Node Coaxial Nanowire (INCN) interconnect technology, and quantifies the system-level performance and energy advantages of the hierarchical icosahedral design when realised on demonstrated 2026 material platforms.

## 2 Geometric Foundation: The Hierarchical Icosahedral Lattice

The Icosahedral Polyhedron Computer is constructed upon a self-similar, hierarchical icosahedral framework. A regular icosahedron possesses 12 vertices, 20 equilateral triangular faces, and 30 edges, exhibiting the full icosahedral symmetry group (Ih) with 5-fold, 3-fold, and 2-fold rotational axes. This symmetry is particularly compatible with the quaternionic and octonionic layers of the Meta-Transistor Engine and with the continuous directional freedom of volumetric spherical logic on the unit Bloch sphere.

The fundamental repeating unit of the architecture is a 13-CNO icosahedral cluster consisting of one central CNO surrounded by twelve outer CNOs positioned at the vertices of a larger icosahedron. Each outer CNO maintains one primary radial connection to the central CNO via a Radial Multi-Bus Nanowire (RMBN) strut and approximately five secondary connections to its nearest neighbouring outer CNOs. This produces a compact, symmetrically connected unit while preserving full icosahedral symmetry. Larger systems are constructed hierarchically by repeating this 13-CNO motif at successive scales, with appropriate golden-ratio scaling between levels to maintain geometric alignment.

The practical interconnect scheme follows a two-tier pattern. Tier 1 consists of twelve radial RMBN connections running from the vertices of the central CNO to the inner vertices of each outer CNO. Tier 2 consists of surface-level connections between neighbouring outer CNOs, with each outer CNO linked to its five nearest neighbours via short RMBN segments. This configuration yields a well-connected network (each outer CNO has six connections in total) while minimising interconnect length and preserving the global icosahedral symmetry. All RMBN segments are engineered to support low-parasitic, phase-preserving transport of the full progressive logic hierarchy, including multi-valued phase and hypercomplex representations.

In the hierarchical realisation, small regular icosahedra (or multi-shell geodesic clusters inspired by buckyonion structures) are positioned at the vertices of larger icosahedral frameworks. Corresponding vertices of adjacent small icosahedra are connected by Radial Multi-Bus Nanowire (RMBN) struts. This produces a fractal “icosahedron of icosahedra” whose every level respects icosahedral symmetry and maps directly onto the radial-shell grading of Hierarchical Shell-Manifold Theory.

The natural interstitial voids that arise in such icosahedral assemblies — which prevent space-filling tessellation without gaps — are intentionally retained and filled with high-thermal-conductivity materials, most notably a structured diamond dodecahedral network. These voids function as engineered thermal pathways that enhance waste-heat extraction and support the supercritical CO<sub>2</sub> recovery systems described in Section 3.4, while the overall structure main-

tains exceptional mechanical robustness and radiation hardness when realised on h-BN/BNNT platforms.

Each vertex of the small icosahedra hosts one or more Meta-Transistor Engine resonant chambers. The RMBN struts along the edges provide low-parasitic, phase-preserving, and probabilistically tunable interconnects that faithfully transport the full progressive logic hierarchy (binary through volumetric spherical and hypercomplex representations) between nodes. The high coordination inherent to the icosahedral symmetry, combined with the multi-layer radial structure of the RMBN, ensures efficient nearest-neighbour connectivity and minimal signal propagation skew despite the presence of intentional thermal voids.

The resulting lattice supports dense three-dimensional arrays, hemispherical and near-spherical focal-plane arrays, and large-area flexible Meta-Skin coverings. The icosahedral geometry provides superior omnidirectional field-of-view coverage and aligns naturally with the polar and hypercomplex encoding native to the Meta-Transistor Engine, enabling accurate source localisation in vision, hearing, and tactile modalities with minimal post-processing.

### 3 The Meta-Transistor Engine: Resonant Leakage Modulator

The fundamental computational element of the Icosahedral Polyhedron Computer is the Meta-Transistor Engine, realised as a high-Q resonant chamber that functions as a controllable HSMT leakage modulator. Each chamber is positioned at a vertex within the hierarchical icosahedral nodes and performs computation, sensing, and non-volatile storage by dynamically tuning the local blurriness field

$$\sigma(r, t) = \sigma_0 - \Delta\sigma \cdot f_{\text{res}}(r, t),$$

where  $\sigma_0 = \sqrt{2}/4$  is the canonical Gaussian kernel width fixed by HSMT and  $f_{\text{res}}(r, t)$  is the normalised resonance response driven by external control tones.

The resonant chamber supports strongly confined optical, mechanical, or hybrid modes with quality factors  $Q > 10^7$ . Computation emerges from the controlled modulation of inter-shell leakage between adjacent HSMT radial shells. By varying the amplitude, frequency, and phase of the drive tones, the chamber can implement the complete logic progression supported by the Compound Multi-Logic Engine (CMLE), including binary, balanced ternary, hidden base-1001, temporal-amplitude encoding, quadrature and arbitrary phase encoding, volumetric spherical logic, native quaternionic and octonionic operations, fractal self-similar logic layers, thermodynamically-aware computation, and an explicit observer/measurement layer.

All operations are performed reversibly through resonance feedback. Information is encoded in the persistent resonance condition (frequency, phase, amplitude, and mode occupation) rather than in charge storage, eliminating the need for conventional volatile memory. The same chamber simultaneously acts as a multimodal sensor: perturbations to the resonant mode induced by optical, acoustic, mechanical, thermal, or chemical stimuli are directly interpreted as computational inputs.

Extensive lattice simulations of the Meta-Transistor Engine have established concrete performance baselines. Deterministic binary gates achieve average switching energies of 0.48 aJ with propagation delays of approximately 2.0 fs. Probabilistic (p-bit) implementations reduce energy consumption by 20–30% while maintaining output fidelity with Kullback–Leibler divergence below 0.03. Ternary gates exhibit roughly 1.8–2.0× higher energy and latency than binary gates but provide significantly higher information density. CMLE instructions involving tier transitions have been simulated with energies between 0.97 aJ and 1.35 aJ per instruction and tier-transition success rates above 98%.

These simulation results are obtained directly from the HSMT radial-lattice model with no additional fitting parameters, confirming that the Meta-Transistor Engine performance is fully determined by the five mathematical constants of HSMT and the material-specific mapping of the resonant chamber.

### 3.1 Overall System Architecture

The Icosahedral Polyhedron Computer is built around three tightly integrated components:

- **Primary Computing Nodes:** Nine-layer hybrid Carbon Nano-Onions ( $\text{Sc}_3\text{N}@\text{C}_{80}$  core with alternating boron nitride and carbon shells). Each carbon shell maintains a strict 6/6 allocation of its twelve pentagons: six dedicated to power delivery using Hybrid RMBNs and six dedicated to data and control signals using standard RMBNs. The multi-layer radial structure and zoned architecture naturally support fractal self-similar logic layers operating across scales, thermodynamically-aware computation, and an observer/measurement layer located primarily in the high-coherence inner zones.
- **Interconnects:** Radial Multi-Bus Nanowires (RMBNs) penetrate each CNO according to the tapering scheme of  $12 \rightarrow 8 \rightarrow 8 \rightarrow 6 \rightarrow 6$  channels. Hybrid RMBNs (approximately 1.0–1.4 nm) are used for power delivery, while standard RMBNs (approximately 0.7–1.0 nm) handle data and control signals. Connections between neighboring CNOs are provided by Inter-Node Coaxial Nanowires (INCNs).
- **Thermal Management:** The natural interstitial voids of the icosahedral lattice are occupied by a structured dodecahedral secondary framework filled with an h-BN/c-BN composite. This network provides both passive heat extraction and the physical foundation for thermodynamically-aware logic operations.

This architecture supports the full advanced capability of the Compound Multi-Logic Engine (CMLE), including fractal self-similar reasoning, thermodynamically-aware computation, and observer-dependent measurement, while maintaining clear separation between power and data pathways throughout the hierarchical structure.

### 3.2 Progressive Logic System and Volumetric Spherical Representation

Building on this integrated architecture, each Meta-Transistor Engine resonant chamber implements the full advanced Compound Multi-Logic Engine (CMLE). This system evolves from binary and balanced ternary through hidden balanced ternary in base-1001, temporal-amplitude encoding, quadrature phase encoding, arbitrary phase encoding via Euler’s formula, and three-axis amplitude with time using the quaternion form of Euler’s formula, ultimately realizing **\*\*volumetric spherical logic\*\*** in which logic digits occupy positions inside the volume of the unit Bloch sphere.

Each resonant chamber encodes its state using a three-component amplitude vector  $\mathbf{A}(t) = (A_x(t), A_y(t), A_z(t))$ . The normalized form

$$\mathbf{r}(t) = \frac{\mathbf{A}(t)}{|\mathbf{A}(t)|}$$

defines a point on the unit Bloch sphere. This representation is grounded in Euler’s formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

which permits continuous and arbitrary phase angles on the amplitude axes. For general three-dimensional rotations of the Bloch vector, the architecture employs the unit quaternion

$$q = \cos \frac{\theta}{2} + \mathbf{u} \sin \frac{\theta}{2}.$$

Because the Meta-Transistor Engine incorporates native quaternionic and octonionic layers, arbitrary orientations and volumetric distributions of logic states within the sphere are supported directly by the underlying algebraic structure.

This continuous directional and volumetric freedom enables the simultaneous, weighted occupation of multiple phase-amplitude states inside the sphere. It therefore provides each resonant chamber with exceptionally high information capacity while remaining fully consistent with the parameter-free foundations of Hierarchical Shell-Manifold Theory.

The CMLE further extends this foundation through **fractal self-similar logic layers** that operate recursively across radial scales, **thermodynamically-aware logic** that distinguishes reversible and irreversible operations, and an **observer/measurement layer** that enables context-dependent readout and reflective computation. These extensions are supported by the multi-scale radial structure and zoned architecture of the nine-layer Carbon Nano-Onion nodes.

### 3.3 Material Platforms and Quantitative Performance Estimates

While the primary physical embodiment of the Meta-Transistor Engine in the Icosahedral Polyhedron Computer is the nine-layer hybrid Carbon Nano-Onion, two alternative material platforms are also considered for the physical realisation of the resonant chambers. Both platforms are compatible with demonstrated 2026 fabrication processes and may be used in specific applications or in hybrid configurations.

**Platform 1: Silicon-on-Lithium-Niobate Photonic-Crystal Defect Cavities** High-Q photonic-crystal cavities are formed in thin-film lithium niobate on silicon. These cavities support strongly confined optical modes with  $Q > 10^7$  and electro-optic tuning bandwidths exceeding 10 GHz. The blurriness field  $\sigma(r, t)$  is modulated via the Pockels effect. This platform is particularly well suited for high-speed deterministic and probabilistic binary logic as well as optical sensing modalities.

**Platform 2: h-BN/BNNT Heterostructure Suspended Membranes** Hexagonal boron nitride membranes encapsulating boron-nitride nanotubes (BNNT) form mechanically robust, high-Q resonators with intrinsic quality factors exceeding  $10^8$  at room temperature. The blurriness field is modulated through electrostatic or optomechanical drive. This platform offers superior radiation hardness and mechanical flexibility, making it ideal for Meta-Skin and harsh-environment applications.

Quantitative performance estimates, derived from radial-lattice simulations and material parameters, are summarised below for both platforms operating at 300 K with optimal doping. These estimates assume conservative quality factors ( $Q = 10^7$ ) and account for interconnect and control overhead. When scaled to a full hierarchical icosahedral lattice with reversible resonant operation using nine-layer hybrid Carbon Nano-Onions, system-level energy reductions of 1,000–3,000 $\times$  relative to conventional CMOS are projected.

| Parameter                                        | Si-on-LN Platform         | h-BN/BNNT Platform        |
|--------------------------------------------------|---------------------------|---------------------------|
| Maximum clock frequency                          | 80–120 GHz                | 40–70 GHz                 |
| Energy per binary gate                           | 0.45–0.55 aJ              | 0.40–0.50 aJ              |
| Energy per ternary gate                          | 0.85–1.05 aJ              | 0.75–0.95 aJ              |
| Energy per CMLE instruction (w/ tier transition) | 1.05–1.35 aJ              | 0.97–1.28 aJ              |
| Operating temperature range                      | 4–400 K                   | 4–600 K                   |
| Radiation hardness (TID)                         | > 1 Mrad                  | > 10 Mrad                 |
| Power density (dense 3D array)                   | 0.8–1.5 W/cm <sup>3</sup> | 0.5–1.2 W/cm <sup>3</sup> |
| Passive retention time (no refresh)              | 1–10 ms                   | 10–100 ms                 |

Table 1: Estimated performance parameters for the two primary material platforms realising the Meta-Transistor Engine in the Icosahedral Polyhedron Computer using nine-layer hybrid Carbon Nano-Onions.

Both platforms support monolithic or heterogeneous integration with the hierarchical icosahedral lattice geometry, enabling flat arrays, hemispherical focal-plane arrays, and flexible Meta-

Skin implementations. The choice between platforms is application-dependent: the silicon-on-lithium-niobate platform favours high-speed, high-density digital and optical processing, while the h-BN/BNNT platform excels in mechanically robust, radiation-hardened, and wide-temperature sensing applications.

### 3.4 Physical Realization: Hybrid Carbon-BN Carbon Nano-Onions

While the Meta-Transistor Engine can be realized in various resonant platforms, one particularly promising physical embodiment uses multi-shell Carbon Nano-Onions (CNOs) with alternating carbon and boron nitride layers. In this hybrid architecture, the CNO begins with an inner  $\text{Sc}_3\text{N@C}_{80}$  core and then alternates between carbon shells (providing conductive resonant layers) and boron nitride shells (providing insulating barriers).

This alternating structure creates well-defined radial zones that align directly with the zoned functional model of the architecture. With nine radial shells, the design provides substantial separation between functional zones. This enables the inner core region to maintain high resonance quality while the middle and outer zones support complex multi-valued, quaternionic, and octonionic operations. The boron nitride layers act as natural physical barriers that enhance electrical isolation between radial levels and contribute to the natural gradient of increasing resonance quality (graduated quietness) toward the center of each CNO.

The hybrid architecture consists of an  $\text{Sc}_3\text{N@C}_{80}$  core followed by eight alternating shells:

$$\text{Sc}_3\text{N@C}_{80}\text{@BN}_{240}\text{@C}_{540}\text{@BN}_{960}\text{@C}_{1500}\text{@BN}_{2160}\text{@C}_{2940}\text{@BN}_{3840}\text{@C}_{4860}$$

Each carbon shell contains twelve pentagons, which serve as the primary attachment points for Radial Multi-Bus Nanowires (RMBNs). These twelve pentagons are allocated in a strict 6/6 configuration that is maintained across all radial layers. The allocation of RMBN channels per layer, following the established tapering scheme, is as follows:

| Layer           | Total RMBNs | Power RMBNs (Hybrid) | Data RMBNs (Standard) |
|-----------------|-------------|----------------------|-----------------------|
| Layer 1 (Outer) | 12          | 6                    | 6                     |
| Layer 2         | 8           | 4                    | 4                     |
| Layer 3         | 8           | 4                    | 4                     |
| Layer 4         | 6           | 3                    | 3                     |
| Layer 5 (Core)  | 6           | 3                    | 3                     |

Table 2: Allocation of RMBN channels per radial layer under the 6/6 pentagon allocation.

Hybrid RMBNs (outer diameter approximately 1.0–1.4 nm) are used on the power-dedicated pentagons, while standard RMBNs (outer diameter approximately 0.7–1.0 nm) are used on the data-dedicated pentagons. Connections between neighboring CNOs in the icosahedral lattice are provided by Inter-Node Coaxial Nanowires (INCNs).

The choice of an  $\text{Sc}_3\text{N@C}_{80}$  core provides a stable, high-symmetry platform with internal multi-orientation states. These internal degrees of freedom support low-tier logic operations while serving as a natural central control region with high resonance quality and coherence.

### 3.5 Capability Mapping to the Compound Multi-Logic Engine (CMLE)

The nine-layer Carbon Nano-Onion architecture maps naturally onto the full advanced capability of the Compound Multi-Logic Engine (CMLE). The substantial radial depth provides excellent separation between functional zones, enabling robust support for the complete progression from binary and balanced ternary through volumetric spherical logic and native quaternionic/octonionic operations.

The multi-scale structure of the CNOs and the hierarchical icosahedral lattice strongly support **fractal self-similar logic layers**, allowing logic operations to function recursively across radial scales. The zoned architecture—with its natural gradient of increasing resonance quality toward the inner core—provides an ideal physical substrate for the **observer/measurement layer**, which can reside primarily in the high-coherence inner zones while interacting with outer computational layers.

The integrated thermal management system (structured diamond dodecahedral network) enables **thermodynamically-aware logic**, allowing the CMLE to distinguish between reversible and irreversible operations and to make energy- and heat-conscious decisions at the machine language level.

With nine radial shells and eight deep aligned pathways (six of which reach the core), the architecture delivers high-efficiency support for demanding hypercomplex and fractal workloads while preserving strong protection of the innermost high-coherence region. This design enables the full CMLE instruction set—including fractal, thermodynamically-aware, and observer-dependent operations—to execute with high fidelity and strong parallel performance.

## 4 Physical Interconnect Architecture: Radial Multi-Bus Nanowires

The functional requirements of the Icosahedral Polyhedron Computer — including level-to-level connectivity across radial shells, graduated quietness, dual-mode operation, and support for volumetric spherical and hypercomplex logic — impose demanding requirements on the physical interconnects. Conventional nanowires are insufficient. To meet these requirements, we introduce the **Radial Multi-Bus Nanowire (RMBN)**, an advanced evolution of the coaxial hybrid nanowire concept.

The RMBN is designed to be implanted through the radial shells of the Carbon Nano-Onions (CNOs), enabling direct, level-specific connections between equivalent radial layers across multiple CNOs in the lattice. This transforms the interconnect from a simple link into a multi-channel radial communication fabric.

### 4.1 Structure of the Radial Multi-Bus Nanowire

Two variants of the Radial Multi-Bus Nanowire (RMBN) are employed within each CNO. The **Hybrid RMBN** (outer diameter approximately 1.0–1.4 nm) is used for power delivery and features a high-conductivity metallic or graphene-based core, a thin boron nitride insulating layer, and a thin carbon outer shell. The **standard RMBN** (outer diameter approximately 0.7–1.0 nm) consists of a carbon nanotube core with a thin boron nitride coating and is used for data and control signal transmission.

Each carbon shell contains twelve pentagons allocated according to a strict 6/6 configuration maintained across all radial layers. The resulting per-layer allocation of RMBN channels, following the tapering scheme of  $12 \rightarrow 8 \rightarrow 8 \rightarrow 6 \rightarrow 6$ , is shown below:

| Layer           | Total RMBNs | Power RMBNs (Hybrid) | Data RMBNs (Standard) |
|-----------------|-------------|----------------------|-----------------------|
| Layer 1 (Outer) | 12          | 6                    | 6                     |
| Layer 2         | 8           | 4                    | 4                     |
| Layer 3         | 8           | 4                    | 4                     |
| Layer 4         | 6           | 3                    | 3                     |
| Layer 5 (Core)  | 6           | 3                    | 3                     |

Table 3: Per-layer allocation of power and data RMBN channels under the locked-in 6/6 pentagon allocation.

Connections **\*\*between\*\*** neighboring CNOs in the icosahedral lattice are handled by separate Inter-Node Coaxial Nanowires (INCNs). This architecture maintains a clear separation between power and data pathways while enabling reliable level-to-level connectivity within each CNO.

## 4.2 Hybrid Material Architecture and Thermal Management

The Icosahedral Polyhedron Computer employs a hybrid material strategy that combines the resonant and logic capabilities of carbon-boron nitride structures with the superior thermal properties of diamond. The primary computing nodes are realised as nine-layer hybrid Carbon Nano-Onions, while the natural interstitial voids of the icosahedral lattice are utilised as engineered thermal management channels. These voids are occupied by a structured dodecahedral network constructed from high-quality diamond.

This hybrid approach provides an effective balance between high thermal conductivity, electrical insulation, and mechanical compatibility with the h-BN/BNNT-based interconnects and CNO platforms. The diamond network makes intimate thermal contact with the outer surfaces of the CNOs and the BNNT sheaths of the Radial Multi-Bus Nanowires, enabling efficient heat extraction while preserving electrical isolation. Waste heat collected by the thermal network is subsequently transferred to supercritical CO<sub>2</sub> power cycles for recovery, with any remaining low-grade heat rejected through dry coolers or passive radiative surfaces.

## 4.3 Secondary Thermal Management: Structured Diamond Dodecahedral Network

A more detailed description of this structured diamond dodecahedral thermal network is provided below. While the primary computing nodes rely on hybrid carbon-boron nitride Carbon Nano-Onions, the interstitial voids of the icosahedral lattice are occupied by a dedicated secondary thermal management structure. This structure takes the form of a self-similar dodecahedral network constructed from high-quality diamond. Rather than filling the voids with bulk diamond, the thermal architecture employs a structured network of diamond nanowires and thin diamond shells arranged according to dodecahedral symmetry.

This design provides several theoretical and practical advantages. Diamond possesses the highest thermal conductivity of any known material, enabling highly efficient extraction of waste heat from the primary resonant chambers. The dodecahedral geometry complements the primary icosahedral lattice, allowing the thermal network to make extensive surface contact with the outer boron nitride layers of the CNOs while maintaining mechanical compatibility. Furthermore, the anisotropic nature of diamond nanowires supports relatively coherent phonon transport, facilitating directional heat flow toward centralized collection points.

Thermal interfaces between the diamond network and the primary CNOs are engineered to minimize boundary resistance. Where necessary, atomically thin transition layers are incorporated to improve phonon matching between diamond and boron nitride. This approach preserves the electrical isolation provided by the outer BN shells of the CNOs while maximizing heat transfer efficiency.

The structured diamond dodecahedral network operates entirely passively, requiring no fluid circulation or moving components. Waste heat collected by the network is subsequently transferred to supercritical CO<sub>2</sub> recovery systems. This architecture ensures that thermal management remains fully consistent with the parameter-free principles of Hierarchical Shell-Manifold Theory while delivering the highest theoretically achievable passive heat extraction performance.

#### 4.4 Waste Heat Management and Recovery

The Meta-Transistor Engine operates at significantly lower power density than conventional CMOS-based systems. The hybrid material architecture — combining BeO or AlN for structural heat spreading, h-BN/BNNT for resonant chambers, and selective CVD diamond — provides excellent internal heat distribution. Waste heat is collected at higher temperatures and converted using a supercritical carbon dioxide (sCO<sub>2</sub>) power cycle. Any remaining low-grade heat is rejected using dry coolers or passive radiative cooling panels. This integrated approach enables high computational density with significantly reduced infrastructure demands.

### 5 Elimination of Volatile Memory Constraints

In conventional computing architectures, volatile memory accounts for a substantial fraction of total system power due to continuous refresh operations. The Icosahedral Polyhedron Computer eliminates this overhead by encoding information directly in the persistent resonance conditions of the Meta-Transistor chambers (frequency, phase, amplitude, and mode occupation). With demonstrated quality factors exceeding  $Q > 10^7$ , individual resonant states can persist for milliseconds to seconds without active refresh. Longer-term persistence is achieved through the Balanced-Base Machine Learning Engine (BBMLE) and slow baseline modulation of resonance conditions.

### 6 Omnidirectional Multimodal Sensing: Hemispherical and Dome FPAs

The hierarchical icosahedral lattice architecture naturally lends itself to the construction of hemispherical and near-spherical dome-shaped focal-plane arrays. The closer-to-spherical geometry of the icosahedral framework provides inherent 360° field-of-view coverage with minimal optical distortion. Each Meta-Transistor Engine pixel functions as a multimodal sensor. The spherical arrangement aligns naturally with the device’s native polar and hypercomplex encoding, enabling accurate source localization in three-dimensional space. On-sensor fusion occurs at the pixel and node level, significantly reducing raw data volume and downstream processing load.

### 7 Meta-Skin: Complete Sensory Covering for Robotics

The Meta-Skin is a thin, flexible, and stretchable sensory-computational covering in which every point on a robot’s surface functions as an active node of the Icosahedral Polyhedron Computer. The underlying architecture is the hierarchical icosahedral lattice fabricated on mechanically compliant substrates. Each Meta-Transistor Engine resonant chamber within the Meta-Skin simultaneously performs multimodal sensing, local computation using the full progressive logic hierarchy, and non-volatile storage via persistent resonance states. The icosahedral geometry and high mechanical flexibility of the h-BN/BNNT platform enable conformal coverage of complex curved surfaces while maintaining continuous nearest-neighbour connectivity and omnidirectional sensing capability.

### 8 Global Power Savings and Energy Impact

The Icosahedral Polyhedron Computer achieves its primary energy advantage by replacing conventional irreversible CMOS switching and volatile memory hierarchies with reversible resonant

operation and non-volatile resonance-state encoding. At the device and gate level, the Meta-Transistor Engine achieves switching energies of 0.40–0.55 aJ for binary gates and 0.75–1.05 aJ for ternary gates. After accounting for interconnect overhead, control electronics, and thermal management, the net system-level energy reduction relative to state-of-the-art 2026 CMOS computing systems is estimated at **300–1,500×**. At global scale, these improvements translate into a reduction of worldwide data-centre and high-performance computing electricity consumption from 1.0–1.5 TW to approximately 1–5 GW — a reduction of two to three orders of magnitude.

## 9 Conclusion

The Icosahedral Polyhedron Computer constitutes a complete physical realisation of Hierarchical Shell-Manifold Theory extended through the Meta-Transistor Engine and the advanced Compound Multi-Logic Engine (CMLE). By arranging high-Q resonant leakage modulators within a hierarchical “icosahedron of icosahedra” lattice with a graduated, dual-mode connectivity scheme, the architecture hosts a progressive, multi-dimensional logic system that evolves from binary and balanced ternary to volumetric spherical logic, augmented by fractal self-similar layers, thermodynamically-aware computation, and an explicit observer/measurement layer.

The nine-layer hybrid Carbon Nano-Onion nodes, interconnected through Radial Multi-Bus Nanowires and Inter-Node Coaxial Nanowires with a strict 6/6 power-data separation, provide both the computational depth and the zoned structure necessary to support these advanced capabilities. Extensive radial-lattice simulations have established concrete performance baselines across silicon-on-lithium-niobate and h-BN/BNNT material platforms.

The architecture preserves every core principle of Hierarchical Shell-Manifold Theory while delivering a concrete, scalable, and energy-efficient engineering realisation. It offers a coherent pathway toward safe, robust, and truly embodied superintelligence that integrates high representational power with physical awareness and reflective computation.

Future work will focus on large-scale experimental validation, compiler and runtime support for fractal and thermodynamically-aware execution, and the integration of these capabilities into complete cognitive architectures.

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